Statistics for Applications

Course Notes

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1 Course Overview

1.1 Goals

The primary objectives of this course are:

- To provide a solid introduction to the mathematical theory behind statistical methods
- To offer theoretical guarantees for the statistical methods used in various applications

1.2 Learning Outcomes

By the end of this class, you will be able to:

1. Formulate real-life situations as statistical problems in mathematical terms
2. Select appropriate statistical methods for your specific problem
3. Understand the implications and limitations of various statistical methods

2 Why Statistics?

Statistics has been essential across diverse fields throughout history. Understanding randomness and making decisions under uncertainty are fundamental challenges that appear in numerous contexts.

2.1 Historical Applications

2.1.1 Hydrology (Netherlands, 10th century)

When building dams and dykes, engineers faced the following problem:

- Dams should be high enough to prevent most floods
- Dams should not be excessively expensive (i.e., not too high)

This requires understanding the *distribution* of flood heights and making probabilistic guarantees.

2.1.2 Insurance

Modern insurance relies heavily on statistical modeling:

- Given a person's driving record, car information, and desired coverage
- What constitutes a fair premium?

2.1.3 Clinical Trials

Medical research depends on rigorous statistical analysis:

- A drug is tested on 100 patients
- 56 patients were cured, 44 showed no improvement
- Question: Is the drug effective, or could these results occur by chance?

2.2 The Common Thread: Randomness

Central Concept

All these examples share a fundamental characteristic: **RANDOMNESS**

Key questions arising from randomness include:

- Notion of average ("fair premium", expected flood height, ...)
- Quantifying chance ("most of the floods", probability of cure, ...)
- Assessing significance and variability

3 Probability vs Statistics

3.1 Probability: Working Forward

Definition 3.1 (Probability)**.** Probability studies randomness when the underlying physical process is completely known.

In some cases, the random mechanism is perfectly understood: dice, cards, roulette, fair coins, etc.

Example

Example 1: Rolling 1 Die
Consider the following game:

- Alice gets 1€ if the number of dots ≥ 3
- Bob gets 2€ if the number of dots ≤ 2
- Question: Who do you want to be: Alice or Bob?

Analysis:

- Alice wins with probability $4/6 = 2/3$, expected value $= 1 \times 2/3 = 0.67$€
- Bob wins with probability $2/6 = 1/3$, expected value $= 2 \times 1/3 = 0.67$€

Both have the same expected value, so the game is fair!

Example

Example 2: Rolling 2 Dice

- Choose a number between 2 and 12
- Win 100€ if you correctly predict the sum of the two dice
- Which number should you choose?

Solution: The sum 7 has the highest probability (6/36), as it can be obtained in the most ways: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1).

Key Insight: In probability, we know the random process (e.g., 1/6 chance for each face, dice are independent), and we *deduce* the probability of outcomes and expected amounts.

3.2 Statistics: Working Backward

Statistics becomes necessary when dealing with more complicated processes where:

- We need to **estimate parameters from data**
- True randomness exists (random sampling, biased coins, measurement error)
- Phenomena are deterministic but too complex to model exactly (statistical modeling)

Fundamental Modeling Principle

$$\textbf{Complicated process} = \textbf{Simple process} + \textbf{Random noise}$$

Good modeling consists of choosing a plausible simple process and an appropriate noise distribution.

3.3 Contrasting Probability and Statistics

Example

Probability Perspective:
"Previous studies showed that the drug was 80% effective. Then we can anticipate that for a study on 100 patients, on average 80 will be cured and at least 65 will be cured with 99.99% probability."

Statistics Perspective:
"We observe that 78/100 patients were cured. We conclude (with appropriate methods) that we are 95% confident that for other studies the drug will be effective on between 69.88% and 86.11% of patients."

Key Distinction:

- **Probability:** Known mechanism → Predict outcomes

- **Statistics:** Observed outcomes → Infer mechanism

4 A Practical Example: The Kissing Study

4.1 Research Question

Study Question

Study Question: "Does a neonatal right-side preference make a surprising romantic reappearance later in life?"

4.2 Study Design

Let p denote the proportion of couples that turn their head to the right when kissing.

Experimental Design:

1. Observe n kissing couples
2. Collect the outcome for each couple (1 for RIGHT, 0 for LEFT)
3. Estimate p with the sample proportion $\hat{p}$

Actual Study:

- Source: "Human behaviour: Adult persistence of head-turning asymmetry" (*Nature*, 2003)
- Sample size: $n = 124$ couples
- Observed: 80 couples turned right
- Sample proportion: $\hat{p} = \frac{80}{124} = 64.5\%$

4.3 Initial Analysis

Observation: 64.5% is notably larger than 50%, suggesting a preference for turning right.

Critical Questions:

- What if our data was RIGHT, RIGHT, LEFT ($n = 3$)? That's 66.7% to the right. Even better?
- Intuitively, we need a large enough sample size n to make reliable conclusions
- How large is "large enough"?

Important Note

Key Question: We need mathematical modeling to understand the accuracy of this estimation procedure.

4.4 Formal Statistical Framework

For $i = 1, \ldots, n$, define random variables:

$$R_i = \begin{cases} 1 & \text{if the } i\text{-th couple turns to the right} \\ 0 & \text{otherwise} \end{cases}$$

The estimator of p is the sample average:

$$\hat{p} = \bar{R}_n = \frac{1}{n} \sum_{i=1}^{n} R_i$$

Question: What is the accuracy of this estimator?

To answer this, we propose a statistical model that describes/approximates the experiment well.

4.5 Statistical Model Assumptions

1. Each R_i is a random variable
2. Each $R_i \sim \text{Ber}(p)$ (Bernoulli distribution with parameter p)
3. $R_1, \ldots, R_n$ are mutually independent

4.6 Discussion of Assumptions

4.6.1 Assumption 1: Randomness

Randomness is a way of modeling lack of information. With perfect information about:

- The conditions of kissing
- What goes on in the kissers' minds
- Physical positioning

Physics or sociology would theoretically allow us to predict the outcome deterministically. Since we lack this information, we model the outcome as random.

4.6.2 Assumption 2: Bernoulli Distribution

Since $R_i \in \{0, 1\}$, each R_i is necessarily a Bernoulli random variable.

More general model: We could assume $R_i \sim \text{Ber}(p_i)$ with different parameters for each couple.

Problem: We don't have enough information to estimate all the p_i's accurately.

Simplification: We assume $p_i = p$ for all i (homogeneity assumption).

4.6.3 Assumption 3: Independence

The independence assumption is reasonable because:

- People were observed at different locations
- Observations occurred at different times
- There's no obvious mechanism for dependence between couples

5 Mathematical Tools for Statistical Inference

5.1 Laws of Large Numbers

Key Theorem

Laws of Large Numbers (Weak and Strong)
Let $X, X_1, X_2, \ldots, X_n$ be i.i.d. random variables with $\mu = \mathbb{E}[X]$ and $\sigma^2 = \mathrm{Var}[X] < \infty$.
Define the sample mean:

$$\bar{X}_n := \frac{1}{n}\sum_{i=1}^{n} X_i$$

Then:

- **Weak LLN:** $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$
- **Strong LLN:** $\bar{X}_n \xrightarrow{\text{a.s.}} \mu$

Interpretation:

- The sample average converges to the true mean
- As n increases, our estimate becomes increasingly accurate
- For the kissing study: $\bar{R}_n \to p$, so $\hat{p}$ is a "consistent" estimator

5.2 The Central Limit Theorem - Detailed Treatment

5.2.1 Formal Statement

Key Theorem

Central Limit Theorem (CLT)
Let $X, X_1, X_2, \ldots, X_n$ be i.i.d. random variables with $\mu = \mathbb{E}[X]$ and $\sigma^2 = \mathrm{Var}[X] < \infty$.
Then:

$$\sqrt{n}\frac{\bar{X}_n - \mu}{\sigma} \xrightarrow{(d)} \mathcal{N}(0,1)$$

Equivalently:

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{(d)} \mathcal{N}(0,\sigma^2)$$

where $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ is the sample mean.

5.2.2 Key Insights and Interpretation

What the CLT Tells Us

1. **Asymptotic Normality:** Regardless of the original distribution of X, the normalized sample mean converges to a normal distribution. This is remarkable because X could follow any distribution (exponential, uniform, Bernoulli, etc.) as long as it has finite variance.

2. **Rate of Convergence:** The difference $(\bar{X}_n - \mu)$ decreases at rate $1/\sqrt{n}$:
   - To halve the error, you need 4 times as many samples
   - To achieve 10× better accuracy, you need 100× more data
   - The standard error is $\sigma/\sqrt{n}$
3. **Universal Result:** The CLT is a universal phenomenon that works for ANY distribution with finite variance. This universality is what makes it so powerful in practice.
4. **Foundation for Inference:** Provides the theoretical basis for:
   - Confidence intervals
   - Hypothesis testing
   - Standard errors and margin of error calculations

Alternative Formulation
The CLT can be rewritten in a more intuitive form. For large n:

$$\bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

Interpretation:

- The sample mean is approximately normally distributed
- Centered at the true mean μ
- With variance that shrinks as $1/n$
- The standard deviation (standard error) is $\sigma/\sqrt{n}$

This approximation becomes more accurate as n increases.

5.2.3 Detailed Application to the Kissing Study

Setup
Recall: $R_i \sim \text{Ber}(p)$ for $i = 1, \ldots, n$, where:

- Mean: $\mu = \mathbb{E}[R_i] = p$
- Variance: $\sigma^2 = \text{Var}[R_i] = p(1-p)$

Applying the CLT
The Central Limit Theorem gives us:

$$\sqrt{n}\frac{\bar{R}_n - p}{\sqrt{p(1-p)}} \xrightarrow{(d)} \mathcal{N}(0,1)$$

Let Φ denote the CDF of the standard normal distribution $\mathcal{N}(0,1)$. For any $x > 0$ and large n:

$$\mathbb{P}\left[\left|\frac{\bar{R}_n - p}{\sqrt{p(1-p)/n}}\right| \leq x\right] \approx 2\Phi(x) - 1$$

Equivalently:

$$\mathbb{P}\left[|\bar{R}_n - p| \leq x\sqrt{\frac{p(1-p)}{n}}\right] \approx 2\Phi(x) - 1$$

Key observation: This tells us how $\bar{R}_n$ concentrates around p. The concentration interval has width proportional to $1/\sqrt{n}$.

5.2.4 Constructing Confidence Intervals from the CLT

General Approach

For a confidence level $1-\alpha$ (where $\alpha \in (0,1)$), let $q_{\alpha/2}$ denote the $(1-\alpha/2)$-quantile of $\mathcal{N}(0,1)$.

Common values:

- For 90% confidence: $\alpha = 0.10$, so $q_{0.05} = 1.645$
- For 95% confidence: $\alpha = 0.05$, so $q_{0.025} = 1.96$
- For 99% confidence: $\alpha = 0.01$, so $q_{0.005} = 2.576$

From the CLT, with probability approximately $1-\alpha$ when n is large:

$$\left|\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}\right| \leq q_{\alpha/2}$$

Rearranging:

$$\mu \in \left[\bar{X}_n - q_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{X}_n + q_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right]$$

Important Note

The Fundamental Problem:
This confidence interval depends on the unknown parameter σ (or p in the Bernoulli case). We cannot use this directly because we don't know the true value of the parameter we're trying to estimate!

Solution 1: Conservative Bounds

For the Bernoulli case, we can use the fact that:

$$p(1-p) \leq \frac{1}{4} \quad \text{for all } p \in [0,1]$$

The maximum occurs at $p = 1/2$. Therefore:

$$\sqrt{\frac{p(1-p)}{n}} \leq \frac{1}{2\sqrt{n}}$$

This gives a conservative confidence interval that holds with probability at least $1-\alpha$:

$$p \in \left[\bar{R}_n - \frac{q_{\alpha/2}}{2\sqrt{n}}, \bar{R}_n + \frac{q_{\alpha/2}}{2\sqrt{n}}\right]$$

Trade-off: This interval is wider than necessary (conservative), but it doesn't depend on the unknown p.

Solution 2: Plug-in Estimator with Slutsky's Theorem
A more efficient approach uses Slutsky's Theorem (discussed in Section 9).
Since $\bar{R}_n \xrightarrow{\mathbb{P}} p$, by the continuous mapping theorem:

$$\sqrt{\bar{R}_n(1-\bar{R}_n)} \xrightarrow{\mathbb{P}} \sqrt{p(1-p)}$$

By Slutsky's Theorem:

$$\sqrt{n}\frac{\bar{R}_n - p}{\sqrt{\bar{R}_n(1-\bar{R}_n)}} \xrightarrow{(d)} \mathcal{N}(0,1)$$

This leads to a narrower, more efficient confidence interval:

$$p \in \left[\bar{R}_n - q_{\alpha/2}\sqrt{\frac{\bar{R}_n(1-\bar{R}_n)}{n}}, \bar{R}_n + q_{\alpha/2}\sqrt{\frac{\bar{R}_n(1-\bar{R}_n)}{n}}\right]$$

5.2.5 Numerical Examples

Example 1: Kissing Study with $n = 124$
Data: $n = 124$ couples, $\hat{p} = 80/124 = 0.645$
For 95% confidence ($\alpha = 0.05$, $q_{0.025} = 1.96$):
Conservative interval:

$$0.645 \pm \frac{1.96}{2\sqrt{124}} = 0.645 \pm 0.088 = [0.557, 0.733]$$

Efficient interval (Slutsky):

$$0.645 \pm 1.96\sqrt{\frac{0.645 \times 0.355}{124}} = 0.645 \pm 0.084 = [0.561, 0.729]$$

Interpretation: We are approximately 95% confident that the true proportion of right-turning couples is between 56.1% and 72.9%. Since the entire interval is above 50%, we have statistical evidence that couples prefer turning right.
Example 2: Warning About Small Samples ($n = 3$)
Data: $n = 3$ couples, $\hat{p} = 2/3 = 0.667$
Naive CLT-based interval:

$$0.667 \pm \frac{1.96}{2\sqrt{3}} = 0.667 \pm 0.566 = [0.101, 1.233]$$

Important Note

Critical Problem: The interval exceeds the valid range $[0, 1]$ for a probability! This demonstrates that the CLT is **not valid for such small samples**. The asymptotic approximation is extremely poor when n is too small.
For small samples, alternative methods must be used:

- Exact binomial confidence intervals
- Hoeffding's inequality (non-asymptotic)
- Bootstrap methods

5.2.6 Requirements and Conditions for the CLT

Essential Requirements

1. **Independence:** The random variables $X_1, \ldots, X_n$ must be mutually independent
2. **Identical Distribution:** All variables must follow the same distribution
3. **Finite Variance:** The variance $\sigma^2 = \text{Var}[X]$ must be finite (i.e., $\sigma^2 < \infty$)

Remark 5.1. These conditions can be relaxed in more advanced versions of the CLT:

- **Lindeberg-Feller CLT:** Allows for non-identical distributions under certain conditions
- **Lyapunov CLT:** Alternative conditions for non-identical distributions
- **Dependent case:** Extensions exist for certain types of dependence (e.g., mixing conditions)

However, the classical CLT with i.i.d. assumptions is the most commonly used in practice.

When is n "Large Enough"?
The CLT is an asymptotic result ($n \to \infty$), but in practice we work with finite samples.
General guidelines:

- **Rule of thumb:** $n \geq 30$ is often cited as sufficient for the CLT approximation
- **Depends on the original distribution:**
  - Symmetric distributions: CLT works well even for smaller n (perhaps $n \geq 15$)
  - Highly skewed distributions: May need $n \geq 50$ or even larger
  - Distributions with heavy tails: Require larger n
  - Nearly normal distributions: CLT approximation good even for small n
- **Bernoulli case:** CLT approximation is reasonable when $np(1-p) \geq 10$

Example

Assessing CLT Validity
For Bernoulli(p) variables:

- If $p = 0.5$ (symmetric): $n = 20$ might suffice
- If $p = 0.1$ or $p = 0.9$ (skewed): need larger n, perhaps $n \geq 50$
- If $p = 0.01$ (highly skewed): need very large n, perhaps $n \geq 100$

The condition $np(1-p) \geq 10$ helps ensure the normal approximation is reasonable.

5.2.7 Visualizing the Central Limit Theorem

Key Visualization Concepts

Consider the distribution of $\bar{X}_n$ for increasing values of n:

- **Small n:** The distribution of $\bar{X}_n$ resembles the original distribution of X
- **Moderate n:** The distribution begins to look more bell-shaped
- **Large n:** The distribution is approximately normal, centered at μ, with variance σ^2/n

Important observations:

1. The center stays at μ for all n
2. The spread decreases as n increases (proportional to $1/\sqrt{n}$)
3. The shape becomes increasingly normal

5.2.8 Connection to Other Fundamental Results

Relationship with Law of Large Numbers

LLN vs CLT

Law of Large Numbers:

- Tells us: $\bar{X}_n \to \mu$ (convergence)
- Answers: "Does the sample mean converge?"
- Type of convergence: Almost sure or in probability

Central Limit Theorem:

- Tells us: How fast and in what manner $\bar{X}_n \to \mu$
- Answers: "How is the sample mean distributed?" and "How fast does it converge?"
- Type of convergence: In distribution (to normal)

Relationship: The CLT provides a refinement of the LLN by quantifying the rate and distributional form of convergence.

Foundation for the Delta Method

The CLT provides the starting point for the **Delta Method** (Section **??**), which extends asymptotic normality to smooth transformations of estimators.

If $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{(d)} \mathcal{N}(0, \sigma^2)$ by the CLT, then for a differentiable function g:

$$\sqrt{n}(g(\bar{X}_n) - g(\mu)) \xrightarrow{(d)} \mathcal{N}(0, [g'(\mu)]^2\sigma^2)$$

This allows us to study nonlinear estimators like $\hat{\lambda} = 1/\bar{X}_n$ in the metro example.

5.2.9 Beyond the Classical CLT: Extensions

Multivariate Central Limit Theorem

For random vectors $\mathbf{X} = (X_1, \ldots, X_d)^T \in \mathbb{R}^d$ with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$:

$$\sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \xrightarrow{(d)} \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$$

This extends all our inference methods to multivariate settings.

Functional Central Limit Theorem

Also known as Donsker's Theorem, this extends the CLT to function spaces, providing the foundation for studying stochastic processes.

Berry-Esseen Theorem

Provides explicit bounds on the rate of convergence in the CLT:

$$\sup_x \left| \mathbb{P}\left[\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \leq x \right] - \Phi(x) \right| \leq \frac{C\mathbb{E}[|X - \mu|^3]}{\sigma^3 \sqrt{n}}$$

where C is a universal constant (approximately 0.4748).

Implication: The error in the CLT approximation decreases at rate $O(1/\sqrt{n})$.

5.2.10 Practical Guidelines for Using the CLT

1. **Check assumptions:**
   - Verify independence (or understand dependence structure)
   - Confirm finite variance
   - Assess whether sample size is adequate
2. **Assess adequacy of normal approximation:**
   - Use diagnostic plots (Q-Q plots, histograms)
   - Consider the shape of the original distribution
   - Apply corrections for small samples if needed
3. **Interpret results correctly:**
   - Confidence intervals have frequentist interpretation
   - The approximation improves with larger n
   - Results are approximate, not exact
4. **Consider alternatives for small samples:**
   - Exact methods (when available)
   - Non-asymptotic concentration inequalities
   - Bootstrap methods
   - Permutation tests

5.2.11 Common Misconceptions

Important Note

Misconception 1: "The CLT says the data becomes normally distributed."
Correction: The CLT says the *sample mean* (or sum) becomes approximately normal. The individual data points retain their original distribution.

Misconception 2: "I need $n \geq 30$ regardless of the distribution."
Correction: The rule $n \geq 30$ is a rough guideline. The required sample size depends heavily on the original distribution's shape.

Misconception 3: "The CLT applies to any statistic."
Correction: The classical CLT specifically applies to sample means. Other statistics may or may not be asymptotically normal (though many are, under appropriate conditions).

Misconception 4: "Larger n always means the CLT works better."
Correction: While true asymptotically, if model assumptions (independence, identical distribution) are violated, increasing n won't fix the problem.

5.2.12 Summary: Why the CLT is Fundamental

The Central Limit Theorem is arguably the most important result in statistics because:

1. **Universality:** Works for any distribution with finite variance
2. **Normality:** Explains why normal distributions appear so frequently in practice
3. **Practical inference:** Provides the foundation for confidence intervals and hypothesis tests
4. **Rate quantification:** Tells us how estimation error decreases with sample size
5. **Asymptotic theory:** Serves as the building block for more sophisticated statistical methods

The CLT, together with the Law of Large Numbers, forms the mathematical foundation for understanding why statistical inference works and provides quantitative tools for practical data analysis.

5.3 Application to the Kissing Study

5.3.1 Consequences of LLN

The Laws of Large Numbers tell us that:

$$\bar{R}_n \xrightarrow{\mathbb{P}} p \quad \text{and} \quad \bar{R}_n \xrightarrow{\text{a.s.}} p$$

Hence, when the sample size n becomes large, $\bar{R}_n$ is a good ("consistent") estimator of p.

5.3.2 Consequences of CLT

Let:

- $\Phi(x) = \text{CDF of } \mathcal{N}(0,1)$
- $\Phi_n(x) = \text{CDF of } \sqrt{n}\frac{\bar{R}_n - p}{\sqrt{p(1-p)}}$

The CLT states: $\Phi_n(x) \approx \Phi(x)$ when n becomes large.
Hence, for all $x > 0$:

$$\mathbb{P}\left[|\bar{R}_n - p| \leq x\sqrt{\frac{p(1-p)}{n}}\right] \approx 2\Phi(x) - 1$$

This provides an approximation of how $\bar{R}_n$ concentrates around p.

5.4 Confidence Intervals

5.4.1 Derivation

For a fixed confidence level $\alpha \in (0,1)$, let $q_{\alpha/2}$ be the $(1-\alpha/2)$-quantile of $\mathcal{N}(0,1)$.
Then, with probability $\approx 1-\alpha$ (when n is large enough):

$$p \in \left[\bar{R}_n - q_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}, \bar{R}_n + q_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}\right]$$

Problem: This interval depends on the unknown parameter p!

5.4.2 Conservative Approach

Note that for any $p \in [0,1]$:

$$p(1-p) \leq \frac{1}{4}$$

The maximum occurs at $p = 1/2$.
Hence, roughly with probability at least $1-\alpha$:

$$p \in \left[\bar{R}_n - \frac{q_{\alpha/2}}{2\sqrt{n}}, \bar{R}_n + \frac{q_{\alpha/2}}{2\sqrt{n}}\right]$$

Definition 5.2 (Asymptotic Confidence Interval)**.** When n becomes large, the interval

$$\left[\bar{R}_n - \frac{q_{\alpha/2}}{2\sqrt{n}}, \bar{R}_n + \frac{q_{\alpha/2}}{2\sqrt{n}}\right]$$

contains p with probability $\approx 1-\alpha$. This is called an **asymptotic confidence interval** for p at level $1-\alpha$.

5.4.3 Numerical Example

For the kissing study with $n = 124$, $\hat{p} = 0.645$, and $\alpha = 0.05$:

- $q_{0.025} = 1.96$ (the 97.5th percentile of standard normal)
- Confidence interval: $0.645 \pm \frac{1.96}{2\sqrt{124}} = [0.56, 0.73]$

Interpretation: We are approximately 95% confident that the true proportion of right-turning couples is between 56% and 73%.

Note: Since the entire interval is above 50%, we have evidence that couples tend to turn right when kissing.

5.4.4 Small Sample Caution

For the extreme case with $n = 3$ and $\hat{p} = 2/3$:

$$0.667 \pm \frac{1.96}{2\sqrt{3}} = [0.10, 1.23]$$

Problem: The CLT is not valid for such small samples! The interval even exceeds 1, which is impossible for a probability.

For small samples, we need exact computations or different methods (e.g., Hoeffding's inequality).

5.5 Hoeffding's Inequality

For situations where n is not large enough for the CLT to be reliable:

Key Theorem

Hoeffding's Inequality (i.i.d. case)
Let n be a positive integer and $X, X_1, \ldots, X_n$ be i.i.d. random variables such that $X \in [a, b]$ almost surely, where $a < b$ are given numbers.
Let $\mu = \mathbb{E}[X]$. Then, for all $\lambda > 0$:

$$\mathbb{P}[|\bar{X}_n - \mu| \geq \lambda] \leq 2e^{-\frac{2n\lambda^2}{(b-a)^2}}$$

5.5.1 Confidence Interval from Hoeffding

For $\alpha \in (0, 1)$, setting the right-hand side equal to α and solving:

$$2e^{-\frac{2n\lambda^2}{(b-a)^2}} = \alpha \implies \lambda = (b-a)\sqrt{\frac{\log(2/\alpha)}{2n}}$$

For Bernoulli variables with $a = 0$ and $b = 1$:

Non-Asymptotic Confidence Interval

With probability $\geq 1 - \alpha$:

$$\bar{R}_n - \sqrt{\frac{\log(2/\alpha)}{2n}} \leq p \leq \bar{R}_n + \sqrt{\frac{\log(2/\alpha)}{2n}}$$

This holds even for small sample sizes n.

Advantages:

- Valid for any sample size n (non-asymptotic)
- Provides probabilistic guarantees without assumptions about normality

Disadvantages:

- Generally more conservative (wider intervals) than CLT-based intervals for large n

6 Types of Convergence

Understanding different modes of convergence is crucial for statistical theory. Different types of convergence provide different guarantees about how random sequences behave.

6.1 Almost Sure Convergence

Definition 6.1 (Almost Sure Convergence)**.** Let $(T_n)_{n\geq 1}$ be a sequence of random variables and T a random variable (which may be deterministic).

We say T_n converges almost surely to T, denoted $T_n \xrightarrow{\text{a.s.}} T$, if:

$$\mathbb{P}[\{\omega : T_n(\omega) \to T(\omega)\}] = 1$$

Interpretation: With probability 1, the sequence $T_n(\omega)$ converges to $T(\omega)$ as a sequence of real numbers for almost every outcome ω in the sample space.

6.2 Convergence in Probability

Definition 6.2 (Convergence in Probability)**.** We say T_n converges in probability to T, denoted $T_n \xrightarrow{\mathbb{P}} T$, if:

$$\mathbb{P}[|T_n - T| \geq \lambda] \to 0 \quad \text{as } n \to \infty, \ \forall \lambda > 0$$

Interpretation: For any tolerance $\lambda > 0$, the probability that T_n deviates from T by more than λ goes to zero as n increases.

6.3 Convergence in L^p

Definition 6.3 (Convergence in L^p)**.** For $p \geq 1$, we say T_n converges in L^p to T, denoted $T_n \xrightarrow{L^p} T$, if:

$$\mathbb{E}[|T_n - T|^p] \to 0 \quad \text{as } n \to \infty$$

Special cases:

- $p = 1$: Convergence in mean
- $p = 2$: Convergence in mean square (most common in statistics)

Interpretation: The p-th moment of the distance between T_n and T goes to zero.

6.4 Convergence in Distribution

Definition 6.4 (Convergence in Distribution)**.** We say T_n converges in distribution to T, denoted $T_n \xrightarrow{(d)} T$, if:

$$\mathbb{P}[T_n \leq x] \to \mathbb{P}[T \leq x] \quad \text{as } n \to \infty$$

for all $x \in \mathbb{R}$ at which the CDF of T is continuous.

Also known as: Weak convergence or convergence in law.

6.5 Characterizations of Convergence in Distribution

Proposition 6.5. The following statements are equivalent:

(i) $T_n \xrightarrow{(d)} T$

(ii) $\mathbb{E}[f(T_n)] \to \mathbb{E}[f(T)]$ for all continuous and bounded functions f

(iii) $\mathbb{E}[e^{ixT_n}] \to \mathbb{E}[e^{ixT}]$ for all $x \in \mathbb{R}$ (characteristic functions converge)

Note: Condition (iii) is particularly useful because characteristic functions uniquely determine distributions.

6.6 Extensions to Random Vectors

Remark 6.6. All these definitions extend naturally to random vectors, i.e., random variables taking values in $\mathbb{R}^d$ for some $d \geq 2$.

6.7 Hierarchy of Convergence

Proposition 6.7 (Relationships Between Modes of Convergence)**.** The following implications hold:

1. If $T_n \xrightarrow{\text{a.s.}} T$, then $T_n \xrightarrow{\mathbb{P}} T$, and the two limits are equal almost surely
2. If $T_n \xrightarrow{L^p} T$, then:
   - $T_n \xrightarrow{L^q} T$ for all $q \leq p$
   - $T_n \xrightarrow{\mathbb{P}} T$
   - The limits are equal almost surely
3. If $T_n \xrightarrow{\mathbb{P}} T$, then $T_n \xrightarrow{(d)} T$

Diagram of implications:

$$\text{a.s.} \Rightarrow \text{in probability} \Rightarrow \text{in distribution}$$

$$L^p \Rightarrow L^q \text{ (for } q \leq p) \Rightarrow \text{in probability}$$

Important: The reverse implications generally do not hold!

6.8 Continuous Mapping Theorem

Theorem 6.8 (Continuous Mapping Theorem)**.** If $f : \mathbb{R} \to \mathbb{R}$ is a continuous function and:

$$T_n \xrightarrow{\text{a.s.}/\mathbb{P}/(d)} T$$

then:

$$f(T_n) \xrightarrow{\text{a.s.}/\mathbb{P}/(d)} f(T)$$

Interpretation: Continuous transformations preserve convergence in all these modes.

6.9 Operations with Limits

6.9.1 Almost Sure and Probability Convergence

Proposition 6.9. If $U_n \xrightarrow{\text{a.s.}/\mathbb{P}} U$ and $V_n \xrightarrow{\text{a.s.}/\mathbb{P}} V$, then:

1. $U_n + V_n \xrightarrow{\text{a.s.}/\mathbb{P}} U + V$
2. $U_n V_n \xrightarrow{\text{a.s.}/\mathbb{P}} UV$
3. If $V \neq 0$ almost surely, then $\frac{U_n}{V_n} \xrightarrow{\text{a.s.}/\mathbb{P}} \frac{U}{V}$

6.9.2 Warning About Convergence in Distribution

Important Note

Important Warning:
In general, these algebraic rules do NOT apply to convergence in distribution unless the pair (U_n, V_n) converges jointly in distribution to (U, V).
For example, if $U_n \xrightarrow{(d)} U$ and $V_n \xrightarrow{(d)} V$ independently, we cannot conclude that $U_n + V_n \xrightarrow{(d)} U + V$ without joint convergence.

Exception: If one of the sequences converges to a constant (see Slutsky's Theorem in Section 9).

7 Another Example: Metro Arrival Times

7.1 Problem Setup

Scenario:

- You observe the times between arrivals of the T (metro) at Kendall station: $T_1, \ldots, T_n$
- You want to model these inter-arrival times
- Your goal is to estimate the rate parameter

7.2 Modeling Assumptions

We assume:

1. $T_1, \ldots, T_n$ are mutually independent
2. Each $T_i \sim \text{Exp}(\lambda)$ (exponential distribution with parameter $\lambda > 0$)
3. All share the same parameter λ

7.3 Discussion of Assumptions

7.3.1 Mutual Independence

Assumption: The inter-arrival times $T_1, \ldots, T_n$ are mutually independent.

Justification: Plausible but not completely verifiable. This is often the case with independence assumptions in practice.

Reality check: Some dependence might exist (e.g., if one train is delayed, the next might also be affected), but the independence assumption is a reasonable first approximation.

7.3.2 Exponential Distribution

Key property: The exponential distribution is *memoryless*:

$$\mathbb{P}[T_1 > t + s \mid T_1 > t] = \mathbb{P}[T_1 > s], \quad \forall s, t \geq 0$$

Interpretation: The remaining waiting time doesn't depend on how long you've already waited.

Additional properties:

- $T_i > 0$ almost surely (waiting times are positive)
- Density function: $f(t) = \lambda e^{-\lambda t}$ for $t \geq 0$
- Mean: $\mathbb{E}[T_1] = \frac{1}{\lambda}$
- Variance: $\text{Var}[T_1] = \frac{1}{\lambda^2}$

7.3.3 Common Parameter

Assumption: All inter-arrival times have the same parameter λ.

Justification: Assumes constant average inter-arrival time.

Limitation: Valid only for limited time periods. Rush hour $\neq$ 11pm service!

For more realistic modeling, one might consider:

- Time-varying rates: $\lambda(t)$
- Non-homogeneous Poisson processes

7.4 Estimation Strategy

Since $\mathbb{E}[T_1] = \frac{1}{\lambda}$, a natural estimate of $\frac{1}{\lambda}$ is the sample mean:

$$\bar{T}_n := \frac{1}{n}\sum_{i=1}^{n} T_i$$

A natural estimator of λ is therefore:

$$\hat{\lambda} := \frac{1}{\bar{T}_n}$$

This is called the **method of moments estimator** (we match sample moments to population moments).

7.5 Asymptotic Properties

7.5.1 Consistency

By the Laws of Large Numbers:

$$\bar{T}_n \xrightarrow{\text{a.s.}/\mathbb{P}} \frac{1}{\lambda}$$

By the continuous mapping theorem (with $g(x) = 1/x$):

$$\hat{\lambda} = \frac{1}{\bar{T}_n} \xrightarrow{\text{a.s.}/\mathbb{P}} \lambda$$

Hence, $\hat{\lambda}$ is a **consistent estimator** of λ.

7.5.2 Asymptotic Normality of Sample Mean

By the Central Limit Theorem:

$$\sqrt{n}\left(\bar{T}_n - \frac{1}{\lambda}\right) \xrightarrow{(d)} \mathcal{N}\left(0, \lambda^{-2}\right)$$

Challenge: How does this transfer to $\hat{\lambda}$? How do we construct confidence intervals for λ?

This requires the **Delta Method**.

8 The Delta Method

The Delta Method is a powerful tool for deriving the asymptotic distribution of transformed estimators.

8.1 Statement of the Delta Method

Key Theorem

Delta Method
Let $(Z_n)_{n\geq 1}$ be a sequence of random variables satisfying:

$$\sqrt{n}(Z_n - \theta) \xrightarrow{(d)} \mathcal{N}(0, \sigma^2)$$

for some $\theta \in \mathbb{R}$ and $\sigma^2 > 0$.
We say $(Z_n)_{n\geq 1}$ is **asymptotically normal around** θ.
Let $g : \mathbb{R} \to \mathbb{R}$ be continuously differentiable at the point θ.
Then:

1. $(g(Z_n))_{n\geq 1}$ is also asymptotically normal
2. More precisely:

$$\sqrt{n}(g(Z_n) - g(\theta)) \xrightarrow{(d)} \mathcal{N}(0, [g'(\theta)]^2\sigma^2)$$

8.2 Intuition

The Delta Method is based on a first-order Taylor expansion:

$$g(Z_n) \approx g(\theta) + g'(\theta)(Z_n - \theta)$$

Multiplying by $\sqrt{n}$:

$$\sqrt{n}(g(Z_n) - g(\theta)) \approx g'(\theta) \cdot \sqrt{n}(Z_n - \theta)$$

Since $\sqrt{n}(Z_n - \theta) \xrightarrow{(d)} \mathcal{N}(0, \sigma^2)$, and multiplying by the constant $g'(\theta)$:

$$g'(\theta) \cdot \sqrt{n}(Z_n - \theta) \xrightarrow{(d)} \mathcal{N}(0, [g'(\theta)]^2\sigma^2)$$

8.3 Application to Metro Example

We have:

$$\sqrt{n}\left(\bar{T}_n - \frac{1}{\lambda}\right) \xrightarrow{(d)} \mathcal{N}\left(0, \frac{1}{\lambda^2}\right)$$

Apply the Delta Method with:

- $Z_n = \bar{T}_n$
- $\theta = \frac{1}{\lambda}$
- $\sigma^2 = \frac{1}{\lambda^2}$
- $g(x) = \frac{1}{x}$, so $g'(x) = -\frac{1}{x^2}$

Therefore:

$$g'\left(\frac{1}{\lambda}\right) = -\frac{1}{(1/\lambda)^2} = -\lambda^2$$

By the Delta Method:

$$\sqrt{n}\left(\hat{\lambda} - \lambda\right) = \sqrt{n}\left(g(\bar{T}_n) - g\left(\frac{1}{\lambda}\right)\right) \xrightarrow{(d)} \mathcal{N}(0, \lambda^4 \cdot \lambda^{-2}) = \mathcal{N}(0, \lambda^2)$$

8.4 Attempting a Confidence Interval

From the asymptotic normality:

$$\sqrt{n}\frac{\hat{\lambda} - \lambda}{\lambda} \xrightarrow{(d)} \mathcal{N}(0, 1)$$

For $\alpha \in (0, 1)$ and large n:

$$\mathbb{P}\left[|\hat{\lambda} - \lambda| \leq \frac{q_{\alpha/2}\lambda}{\sqrt{n}}\right] \approx 1 - \alpha$$

Important Note

Problem: The interval

$$\left[\hat{\lambda} - \frac{q_{\alpha/2}\lambda}{\sqrt{n}}, \hat{\lambda} + \frac{q_{\alpha/2}\lambda}{\sqrt{n}}\right]$$

depends on the unknown parameter λ!
This cannot be used as a confidence interval because we don't know λ (that's what we're trying to estimate).

8.5 Two Solutions

8.5.1 Solution 1: Algebraic Manipulation

Starting from:

$$|\hat{\lambda} - \lambda| \leq \frac{q_{\alpha/2}\lambda}{\sqrt{n}}$$

We can rearrange:

$$\begin{aligned}
-\frac{q_{\alpha/2}\lambda}{\sqrt{n}} \leq \hat{\lambda} - \lambda &\leq \frac{q_{\alpha/2}\lambda}{\sqrt{n}} \\
\hat{\lambda} - \frac{q_{\alpha/2}\lambda}{\sqrt{n}} \leq \lambda &\leq \hat{\lambda} + \frac{q_{\alpha/2}\lambda}{\sqrt{n}} \\
\lambda\left(1 - \frac{q_{\alpha/2}}{\sqrt{n}}\right) \leq \hat{\lambda} &\leq \lambda\left(1 + \frac{q_{\alpha/2}}{\sqrt{n}}\right)
\end{aligned}$$

Solving for λ:

$$\hat{\lambda}\left(1 + \frac{q_{\alpha/2}}{\sqrt{n}}\right)^{-1} \leq \lambda \leq \hat{\lambda}\left(1 - \frac{q_{\alpha/2}}{\sqrt{n}}\right)^{-1}$$

Confidence Interval (Method 1)

An asymptotic $(1-\alpha)$ confidence interval for λ is:

$$\left[\hat{\lambda}\left(1+\frac{q_{\alpha/2}}{\sqrt{n}}\right)^{-1}, \hat{\lambda}\left(1-\frac{q_{\alpha/2}}{\sqrt{n}}\right)^{-1}\right]$$

8.5.2 Solution 2: Slutsky's Theorem

A more systematic approach uses Slutsky's Theorem (see next section).

9 Slutsky's Theorem

Slutsky's Theorem provides a systematic way to handle unknown parameters in asymptotic distributions.

9.1 Statement

Key Theorem

Slutsky's Theorem
Let (X_n) and (Y_n) be two sequences of random variables such that:

(i) $X_n \xrightarrow{(d)} X$

(ii) $Y_n \xrightarrow{\mathbb{P}} c$

where X is a random variable and c is a constant.
Then:

$$(X_n, Y_n) \xrightarrow{(d)} (X, c)$$

In particular:

- $X_n + Y_n \xrightarrow{(d)} X + c$
- $X_n Y_n \xrightarrow{(d)} cX$
- If $c \neq 0$, then $\frac{X_n}{Y_n} \xrightarrow{(d)} \frac{X}{c}$

Key Insight: You can replace a sequence converging in probability to a constant with that constant in expressions involving convergence in distribution.

9.2 Application to Metro Example

Step 1: From the Delta Method, we know:

$$\sqrt{n}\frac{\hat{\lambda}-\lambda}{\lambda} \xrightarrow{(d)} \mathcal{N}(0,1)$$

Step 2: By the weak Law of Large Numbers and continuous mapping:

$$\hat{\lambda} \xrightarrow{\mathbb{P}} \lambda$$

Step 3: Apply Slutsky's Theorem:

- Let $X_n = \sqrt{n}\frac{\hat{\lambda}-\lambda}{\lambda} \xrightarrow{(d)} \mathcal{N}(0,1)$
- Let $Y_n = \frac{\lambda}{\hat{\lambda}} \xrightarrow{\mathbb{P}} 1$

Then:

$$X_n \cdot Y_n = \sqrt{n}\frac{\hat{\lambda}-\lambda}{\lambda} \cdot \frac{\lambda}{\hat{\lambda}} = \sqrt{n}\frac{\hat{\lambda}-\lambda}{\hat{\lambda}} \xrightarrow{(d)} \mathcal{N}(0,1)$$

9.3 Confidence Interval Using Slutsky

From the result above, for large n:

$$\mathbb{P}\left[\left|\sqrt{n}\frac{\hat{\lambda}-\lambda}{\hat{\lambda}}\right| \leq q_{\alpha/2}\right] \approx 1-\alpha$$

This gives:

$$\mathbb{P}\left[|\hat{\lambda}-\lambda| \leq \frac{q_{\alpha/2}\hat{\lambda}}{\sqrt{n}}\right] \approx 1-\alpha$$

Confidence Interval (Method 2)

Another asymptotic $(1-\alpha)$ confidence interval for λ is:

$$\left[\hat{\lambda} - \frac{q_{\alpha/2}\hat{\lambda}}{\sqrt{n}}, \hat{\lambda} + \frac{q_{\alpha/2}\hat{\lambda}}{\sqrt{n}}\right]$$

Equivalently:

$$\hat{\lambda}\left[1 - \frac{q_{\alpha/2}}{\sqrt{n}}, 1 + \frac{q_{\alpha/2}}{\sqrt{n}}\right]$$

Advantage: This interval only depends on the observed estimate $\hat{\lambda}$, not on the unknown λ.

9.4 Revisiting the Kissing Study

Recall: In the kissing study, we used the trick that $p(1-p) \leq 1/4$ to construct a confidence interval.

Alternative using Slutsky:

We know:

$$\sqrt{n}\frac{\bar{R}_n - p}{\sqrt{p(1-p)}} \xrightarrow{(d)} \mathcal{N}(0,1)$$

Since $\bar{R}_n \xrightarrow{\mathbb{P}} p$, by continuous mapping:

$$\sqrt{\bar{R}_n(1-\bar{R}_n)} \xrightarrow{\mathbb{P}} \sqrt{p(1-p)}$$

By Slutsky's Theorem:

$$\sqrt{n}\frac{\bar{R}_n - p}{\sqrt{\bar{R}_n(1-\bar{R}_n)}} \xrightarrow{(d)} \mathcal{N}(0,1)$$

Alternative Confidence Interval for p

An asymptotic $(1-\alpha)$ confidence interval for p is:

$$\left[\bar{R}_n - q_{\alpha/2}\sqrt{\frac{\bar{R}_n(1-\bar{R}_n)}{n}}, \bar{R}_n + q_{\alpha/2}\sqrt{\frac{\bar{R}_n(1-\bar{R}_n)}{n}}\right]$$

Comparison:

- The interval using $p(1-p) \leq 1/4$ is more conservative (wider)
- The interval using Slutsky is typically narrower and more efficient
- Both are asymptotically valid

10 Summary and Key Takeaways

10.1 Statistical Thinking Framework

The general approach to statistical problems:

1. Start with a real-world problem
2. Formulate a mathematical/statistical model with reasonable assumptions
3. Apply appropriate statistical methods (estimation, testing, etc.)
4. Interpret results in the context of the original problem
5. Acknowledge limitations and assumptions

10.2 Fundamental Distinction: Probability vs. Statistics

Probability	**Statistics**
Known mechanism	Unknown mechanism
Predict outcomes	Infer mechanism
Deductive reasoning	Inductive reasoning
$\mathbb{P}[\text{data} \mid \text{model}]$	$\mathbb{P}[\text{model} \mid \text{data}]$

10.3 Essential Mathematical Tools

1. **Law of Large Numbers:**
   - Guarantees consistency of sample averages
   - $\bar{X}_n \to \mu$ as $n \to \infty$

2. **Central Limit Theorem:**
   - Quantifies rate of convergence
   - Provides asymptotic normality
   - Foundation for confidence intervals
3. **Hoeffding's Inequality:**
   - Non-asymptotic concentration bounds
   - Valid for finite samples
   - More conservative than CLT-based methods
4. **Delta Method:**
   - Transforms asymptotic distributions
   - Based on first-order Taylor expansion
   - Essential for nonlinear estimators
5. **Slutsky's Theorem:**
   - Handles unknown parameters systematically
   - Allows replacement of consistent estimators
   - Key tool for constructing confidence intervals

10.4 Convergence Concepts

Understanding the hierarchy:

$$\text{Almost sure} \Rightarrow \text{In probability} \Rightarrow \text{In distribution}$$

Key points:

- Different modes provide different guarantees
- Continuous transformations preserve convergence
- Algebraic operations work for a.s. and probability convergence
- Special care needed for convergence in distribution (use Slutsky!)

10.5 Confidence Intervals

General strategy for constructing asymptotic confidence intervals:

1. Establish consistency: $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta$
2. Derive asymptotic normality: $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{(d)} \mathcal{N}(0, \sigma^2)$
3. Handle unknown parameters using:
   - Conservative bounds (e.g., $p(1-p) \leq 1/4$)
   - Algebraic manipulation
   - Slutsky's Theorem (most systematic)
4. Construct interval of the form: $\hat{\theta}_n \pm \frac{q_{\alpha/2}\hat{\sigma}}{\sqrt{n}}$

10.6 Practical Considerations

Model assumptions matter:

- Independence assumptions are often unverifiable but necessary
- Distributional assumptions (e.g., exponential, Bernoulli) should be justified
- Homogeneity assumptions (same parameter) are simplifications

Sample size considerations:

- CLT is asymptotic: requires "large enough" n
- Rule of thumb: $n \geq 30$ for CLT to be reasonable
- For small samples, use exact methods or concentration inequalities

Interpretation of confidence intervals:

- Frequentist interpretation: The procedure has coverage probability $1 - \alpha$
- NOT: "The probability that θ is in this interval is $1 - \alpha$"
- The parameter θ is fixed; the interval is random

10.7 Next Steps

These foundational concepts prepare you for:

- Hypothesis testing
- Linear regression and generalized linear models
- Maximum likelihood estimation
- Bayesian inference
- Non-parametric methods
- High-dimensional statistics

Always remember: Statistical methods provide powerful tools, but their validity depends critically on:

1. Appropriateness of model assumptions
2. Adequate sample size
3. Proper interpretation of results
4. Understanding of limitations

Additional Resources

Recommended Textbooks

- **Mathematical Statistics:**
  - Wasserman, L. (2004). *All of Statistics*. Springer.
  - van der Vaart, A. W. (1998). *Asymptotic Statistics*. Cambridge University Press.
- **Probability Theory:**
  - Billingsley, P. (1995). *Probability and Measure*. Wiley.
  - Durrett, R. (2019). *Probability: Theory and Examples*. Cambridge University Press.
- **Applied Statistics:**
  - Casella, G., & Berger, R. L. (2002). *Statistical Inference*. Duxbury.
  - Wackerly, D., Mendenhall, W., & Scheaffer, R. L. (2014). *Mathematical Statistics with Applications*. Cengage.

Online Resources

- MIT OpenCourseWare: 18.650 Statistics for Applications
- Stanford Online: Statistical Learning
- Coursera: Mathematical Statistics specializations